



# Methods

## Statistical analysis–Linear mixed-effects regression

- Linear mixed-effects models (LMMs) were fitted to predict distinctiveness.
  - Predictors were surprisal, phone1/phone2 flexibility, distinctiveness type (articulatory vs. acoustic), phone2 frequency and phone2 duration.
  - Random intercepts/slopes on surprisal were grouped by speaker, word, and bigram.
- Bigrams that crossed syllable boundaries were excluded.





# Methods

## Statistical analysis–Limited imaging

- Compared to MRI, ultrasound & EMA are limited in terms of the articulators that could be imaged:
  - Ultrasound: excluded /ɹ/, /m, n, ŋ/, /ɑ, ɒ, ɐ, æ, a, aʊ, aɪ, ə/ (56.37%)
  - EMA: excluded /m, n, ŋ/ (19.53%)